

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

CHAPTER 9: ELECTROCHEMISTRY

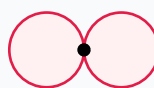
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Electrochemistry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



p orbital (px)

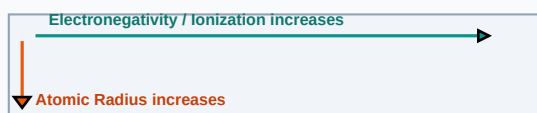
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Molar Conductivity	The conducting power of all the ions produced by dissolving one mole of electrolyte in solution.
Kohlrausch Law	At infinite dilution, the limiting molar conductivity of an electrolyte is the sum of individual ionic contributions.
Standard Hydrogen Electrode (SHE)	A reference electrode containing a platinum plate in 1M H ⁺ gas-purged solution, defined as 0.00V.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

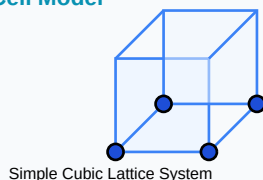
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Molar Conductivity: $\Lambda_m = (\kappa \times 1000) / M$ (κ = specific conductance)
- Nernst Equation: $E_{\text{cell}} = E_{0_{\text{cell}}} - (0.0591 / n) \times \log_{10}(Q)$ at 298K
- Gibbs energy: $\Delta G = -n \times F \times E_{\text{cell}}$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

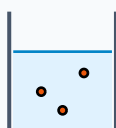
Example 1: Calculate cell potential of Daniell cell at 25C when $\text{Zn}^{2+}=0.1\text{M}$, $\text{Cu}^{2+}=1.0\text{M}$. ($E_0 = 1.10\text{V}$)

Step-by-step Solution:

Reaction: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$. $n = 2$. $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 0.1$.

$E = E_0 - (0.0591 / 2) * \log_{10}(0.1) = 1.10 - 0.0295 * (-1) = 1.10 + 0.0295 = 1.13\text{V}$.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: State Kohlrausch's law and explain how it evaluates limiting molar conductivity of weak electrolytes like Acetic acid.

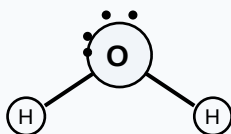
Analytical Solution Strategy:

Kohlrausch states limiting conductivity is sum of ion conductivities.

For CH_3COOH : $\Lambda_0(\text{CH}_3\text{COOH}) = \lambda_0(\text{CH}_3\text{COO}^-) + \lambda_0(\text{H}^+)$.

Evaluated by combining strong electrolytes: $\Lambda_0(\text{CH}_3\text{COONa}) + \Lambda_0(\text{HCl}) - \Lambda_0(\text{NaCl})$.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

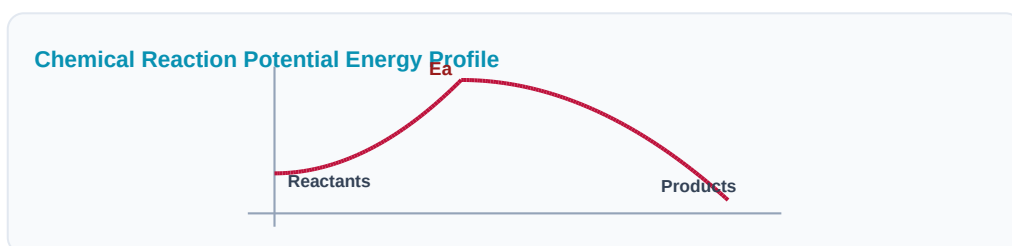
Q1 (1-Mark): Define specific conductance and cell constant.

Q2 (2-Mark): State Faraday's two laws of electrolysis.

Q3 (2-Mark): Describe the construction and working of a Daniell Cell.

Q4 (5-Mark): Write the Nernst equation and define each term.

Q5 (5-Mark): Explain electrochemical rusting corrosion.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Heavy Smelting

Electrolytic refineries isolate structural copper using copper sulfate solutions.

Application Case: Automotive Systems

Lead-acid battery grids store electric power via reversible chemical transfers.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

